

CBSE Class 12 Maths — Continuity, Differentiability and Application of Derivatives

50 Most-Probable PYQs & Board-Pattern Practice Questions

1-Mark (MCQ/Assertion-Reason/VSA)

Q1. Find the derivative of $\sin^{-1}(2x\sqrt{1-x^2})$ with respect to x , assuming $-\frac{1}{\sqrt{2}} < x < \frac{1}{\sqrt{2}}$.

[PYQ 2020] [Confidence: Very High]

Q2. Check whether the function defined by $f(x) = \cos(x^2)$ is continuous at all $x \in \mathbb{R}$. [PYQ 2018] [Confidence: High]

Q3. What is the slope of the normal to the curve $y = 2x^2 + 3 \sin x$ at $x = 0$? [PYQ 2019] [Confidence: High]

Q4. The radius of a circle is increasing uniformly at the rate of **3 cm/s**. What is the rate of change of the area of the circle with respect to time when the radius is **6 cm**? [PYQ 2023] [Confidence: Very High]

Q5. What is the derivative of $e^{\sin^{-1} x}$ with respect to x ? [Practice] [Confidence: Medium]

Q6. Find the derivative of $\log_e(\cos e^x)$ with respect to x . [PYQ 2017] [Confidence: High]

Q7. Assertion (A): The function $f(x) = |x - 1|$ is continuous at $x = 1$ but not differentiable at $x = 1$. **Reason (R):** The left-hand derivative and right-hand derivative of $f(x)$ at $x = 1$ exist but are not equal to each other. Choose the correct option. [PYQ 2024] [Confidence: Very High]

Q8. For what interval is the function $y = x^2 e^{-x}$ strictly increasing? [PYQ 2020] [Confidence: Very High]

Q9. What is the derivative of a^x with respect to x , where $a > 0$ and $a \neq 1$? [Practice] [Confidence: High]

Q10. Differentiate $\sin^2 x$ with respect to $\cos^2 x$. [PYQ 2016] [Confidence: High]

Q11. Find the slope of the tangent to the curve $y = x^3 - 3x$ at the point $x = 2$. [Practice] [Confidence: Medium]

Q12. Find the value of constant k so that the function $f(x) = kx + 1$ for $x \leq \pi$ and $f(x) = \cos x$ for $x > \pi$ is continuous at $x = \pi$. [PYQ 2022] [Confidence: Very High]

Q13. Find the derivative of $\log_e(\log_e x)$, where $x > 1$. [PYQ 2018] [Confidence: Medium]

Q14. What is the absolute maximum value of the function $f(x) = \sin x + \cos x$ in the interval $[0, \pi]$? [PYQ 2019] [Confidence: High]

Q15. The radius of a spherical soap bubble is increasing at the rate of **0.2 cm/s**. What is the rate of increase of its volume when the radius is **5 cm**? [PYQ 2023] [Confidence: Very High]

Q16. Find $\frac{dy}{dx}$ if $x = at^2$ and $y = 2at$. [PYQ 2017] [Confidence: High]

Q17. The function $f(x) = x^3 - 3x^2 + 3x - 100$ is strictly increasing on $\mathbb{R}$. (True/False) [PYQ 2021] [Confidence: High]

Q18. Assertion (A): The function $f(x) = e^x$ is strictly increasing on $\mathbb{R}$. **Reason (R):** A function $f(x)$ is strictly increasing on an interval if $f'(x) > 0$ for all x in that interval. Choose the correct option. [PYQ 2024] [Confidence: High]

Q19. Assertion (A): The function $f(x) = |x|$ is continuous everywhere but not differentiable at $x = 0$. **Reason (R):** Every continuous function is automatically a differentiable function. Choose the correct option. [Practice] [Confidence: High]

Q20. Find the derivative of $2^{\sin x}$ with respect to x . [Practice] [Confidence: Medium]

2-Mark

Q21. Differentiate $y = x^x$ with respect to x . [PYQ 2018] [Confidence: High]

Q22. If $y = \sqrt{\sin x + y}$, find $\frac{dy}{dx}$. [PYQ 2019] [Confidence: Very High]

Q23. Find $\frac{dy}{dx}$ if $y = \cos^{-1} \left(\frac{1-x^2}{1+x^2} \right)$, where $0 < x < 1$. [PYQ 2020] [Confidence: Very High]

Q24. The volume of a cube is increasing at a rate of $8 \text{ cm}^3/\text{s}$. How fast is the surface area increasing when the length of an edge is 12 cm ? [PYQ 2022] [Confidence: Very High]

Q25. If $y = A \sin x + B \cos x$, prove that $\frac{d^2y}{dx^2} + y = 0$. [PYQ 2017] [Confidence: High]

Q26. Find the intervals in which the function $f(x) = x^2 - 4x + 6$ is strictly increasing. [Practice] [Confidence: Medium]

Q27. A ladder 5 m long is leaning against a vertical wall. The bottom of the ladder is pulled along the ground, away from the wall, at the rate of 2 cm/s . How fast is its height on the wall decreasing when the foot of the ladder is 4 m away from the wall? [PYQ 2023] [Confidence: Very High]

Q28. If $x^y = y^x$, find $\frac{dy}{dx}$. [PYQ 2016] [Confidence: High]

Q29. Find $\frac{dy}{dx}$ if $y = x^{\sin x}$. [PYQ 2021] [Confidence: High]

Q30. Find the rate of change of the area of a circle with respect to its radius r when $r = 5 \text{ cm}$. [Practice] [Confidence: Medium]

Q31. If $y = \log(x + \sqrt{x^2 + a^2})$, prove that $\frac{dy}{dx} = \frac{1}{\sqrt{x^2 + a^2}}$. [PYQ 2018] [Confidence: High]

Q32. Prove that the curves $x = y^2$ and $xy = k$ intersect orthogonally (cut at right angles) if $8k^2 = 1$. [PYQ 2020] [Confidence: Very High]

3-Mark

Q33. Differentiate $y = x^x + (\sin x)^{\log x}$ with respect to x . [PYQ 2022] [Confidence: Very High]

Q34. Find the equation of the tangent and normal to the curve $y = x^3 - 3x + 5$ at the point $(2, 7)$. [PYQ 2019] [Confidence: High]

Q35. Verify Rolle's Theorem for the function $f(x) = x^2 + 2x - 8$ on $[-4, 2]$. [Practice] [Confidence: Medium]

Q36. If $x = a(\theta - \sin \theta)$ and $y = a(1 + \cos \theta)$, find the value of $\frac{dy}{dx}$ at $\theta = \frac{\pi}{3}$. [PYQ 2018]

[Confidence: Very High]

Q37. Differentiate $\tan^{-1} \left(\frac{\sqrt{1+x^2}-1}{x} \right)$ with respect to $\tan^{-1} x$. [PYQ 2017] [Confidence: High]

Q38. If $y = (\sin^{-1} x)^2$, prove that $(1 - x^2) \frac{d^2y}{dx^2} - x \frac{dy}{dx} - 2 = 0$. [PYQ 2023] [Confidence: Very High]

Q39. Find the absolute maximum and absolute minimum values of the function $f(x) = 3x^4 - 8x^3 + 12x^2 - 48x + 25$ on the interval $[1]$. [PYQ 2020] [Confidence: High]

Q40. Find the intervals in which the function $f(x) = \sin x + \cos x$, $0 \leq x \leq 2\pi$, is strictly increasing or strictly decreasing. [PYQ 2019] [Confidence: High]

Q41. Use logarithmic differentiation to find $\frac{dy}{dx}$ if $y = x^{\cos x} + (\cos x)^x$. [PYQ 2021] [Confidence: Very High]

Q42. Show that $y = \log(1+x) - \frac{2x}{2+x}$, for $x > -1$, is an increasing function of x throughout its domain. [PYQ 2018] [Confidence: High]

5-Mark (Long Answer/Case-Study)

Q43. Show that the semi-vertical angle of a right circular cone of maximum volume and of a given slant height is $\tan^{-1} \sqrt{2}$. [PYQ 2020] [Confidence: Very High]

Q44. Find the intervals in which the function $f(x) = 2x^3 - 3x^2 - 36x + 7$ is strictly increasing or strictly decreasing. Also, find the coordinates of the points of local maxima and local minima. [PYQ 2023] [Confidence: Very High]

Q45. Show that the altitude of the right circular cylinder of maximum volume that can be inscribed in a given sphere of radius R is $\frac{2R}{\sqrt{3}}$. [PYQ 2019] [Confidence: Very High]

Q46. An open box with a square base is to be made out of a given quantity of cardboard of area c^2 square units. Show that the maximum volume of the box is $\frac{c^3}{6\sqrt{3}}$ cubic units. [PYQ 2018] [Confidence: High]

Q47. A wire of length **28 m** is to be cut into two pieces. One of the pieces is to be bent into the shape of a square and the other into the shape of a circle. What should be the exact length of the two pieces so that the combined area of the square and the circle is minimum? [PYQ 2022] [Confidence: Very High]

Q48. Find the coordinates of the points of local maxima and local minima for the function $f(x) = \frac{3}{4}x^4 - 8x^3 + \frac{45}{2}x^2 - 105$. [Practice] [Confidence: Medium]

Q49. Case Study: A telephone company in a town has 500 subscribers on its list and currently collects fixed charges of ₹300 per subscriber per year. The company proposes to increase the annual subscription to generate more profit. Market research shows that for every increase of ₹1 in the annual subscription, exactly one subscriber will discontinue the service. Let x be the increase in the annual subscription fee in rupees. (i) Write the total revenue function $R(x)$ of the company in terms of x . (1 Mark) (ii) Find $R'(x)$ and equate it to zero to find the critical point. (2 Marks) (iii) Use the second derivative test to find the increase in subscription x that yields the

maximum revenue, and calculate this maximum revenue. (2 Marks) [PYQ 2024] [Confidence: Very High]

Q50. Case Study: An architect is designing a stadium window in the shape of a rectangle surmounted by a semi-circle. To maintain structural integrity and aesthetics, the total perimeter of the window is fixed at exactly **10 meters**. Let x be the width of the rectangular part (which also serves as the diameter of the semi-circle) and y be the height of the rectangular part. (i) Express the perimeter equation in terms of x and y , and write y in terms of x . (1 Mark) (ii) Express the total area A of the window purely as a function of x . (2 Marks) (iii) Find the dimensions x and y that will maximize the area of the window to let in the maximum possible amount of light. (2 Marks) [Practice] [Confidence: High]

Answer Key

Q1. $y' = \frac{2}{\sqrt{1-x^2}}$. **Q2.** Yes, it is continuous everywhere (composition of two continuous functions x^2 and $\cos x$). **Q3.** $\frac{dy}{dx} = 4x + 3 \cos x$. At $x = 0$, slope of tangent is 3. Slope of normal = $-1/3$. **Q4.** $A = \pi r^2 \Rightarrow \frac{dA}{dt} = 2\pi r \frac{dr}{dt} = 2\pi(6)(3) = 36\pi \text{ cm}^2/\text{s}$. **Q5.** $\frac{dy}{dx} = \frac{e^{\sin^{-1}x}}{\sqrt{1-x^2}}$. **Q6.** $\frac{dy}{dx} = -e^x \tan(e^x)$. **Q7.** Option A (Both A and R are true, and R is the correct explanation of A). **Q8.** $y' = e^{-x}(2x - x^2)$. Strictly increasing when $x(2 - x) > 0$, i.e., $x \in (0, 2)$. **Q9.** $\frac{dy}{dx} = a^x \log_e a$. **Q10.** Let $u = \sin^2 x$, $v = \cos^2 x$. $u + v = 1 \Rightarrow \frac{du}{dv} = -1$. **Q11.** $\frac{dy}{dx} = 3x^2 - 3$. At $x = 2$, slope is $3(4) - 3 = 9$. **Q12.** LHL = $k\pi + 1$, RHL = $\cos \pi = -1$. $k\pi + 1 = -1 \Rightarrow k = -2/\pi$. **Q13.** $\frac{dy}{dx} = \frac{1}{x \log_e x}$. **Q14.** $f'(x) = \cos x - \sin x = 0 \Rightarrow x = \pi/4$. $f(\pi/4) = \sqrt{2}$. Endpoints $f(0) = 1$, $f(\pi) = -1$. Absolute max is $\sqrt{2}$. **Q15.** $V = \frac{4}{3}\pi r^3 \Rightarrow \frac{dV}{dt} = 4\pi r^2 \frac{dr}{dt} = 4\pi(25)(0.2) = 20\pi \text{ cm}^3/\text{s}$. **Q16.** $\frac{dx}{dt} = 2at$, $\frac{dy}{dt} = 2a \Rightarrow \frac{dy}{dx} = \frac{1}{t}$. **Q17.** True ($f'(x) = 3(x-1)^2 \geq 0$, and vanishes only at discrete point $x = 1$). **Q18.** Option A (Both A and R are true, and R is the correct explanation of A). **Q19.** Option C (A is true, but R is false; continuous functions are not always differentiable). **Q20.** $\frac{dy}{dx} = 2^{\sin x} \cdot \cos x \cdot \log_e 2$. **Q21.** Take $\log y = x \log x \Rightarrow \frac{dy}{dx} = x^x(1 + \log x)$. **Q22.** $y^2 = \sin x + y \Rightarrow 2yy' = \cos x + y' \Rightarrow \frac{dy}{dx} = \frac{\cos x}{2y-1}$. **Q23.** $y = 2 \tan^{-1} x \Rightarrow \frac{dy}{dx} = \frac{2}{1+x^2}$. **Q24.** $V = a^3 \Rightarrow 3a^2 \frac{da}{dt} = 8 \Rightarrow \frac{da}{dt} = \frac{8}{3a^2}$. Surface area $S = 6a^2 \Rightarrow \frac{dS}{dt} = 12a(\frac{8}{3a^2}) = \frac{32}{a}$. At $a = 12$, $\frac{dS}{dt} = \frac{32}{12} = \frac{8}{3} \text{ cm}^2/\text{s}$. **Q25.** $y' = A \cos x - B \sin x$. $y'' = -A \sin x - B \cos x = -y$. Thus $y'' + y = 0$. **Q26.** $f'(x) = 2x - 4 > 0 \Rightarrow x > 2$. Interval is $(2, \infty)$. **Q27.** $x^2 + y^2 = 25 \Rightarrow 2x \frac{dx}{dt} + 2y \frac{dy}{dt} = 0$. When $x = 4$, $y = 3$. $4(2) + 3 \frac{dy}{dt} = 0 \Rightarrow \frac{dy}{dt} = -\frac{8}{3} \text{ cm/s}$. **Q28.** $y \log x = x \log y \Rightarrow \frac{y}{x} + \log x \cdot y' = \log y + \frac{x}{y} \cdot y' \Rightarrow \frac{dy}{dx} = \frac{y(x \log y - y)}{x(y \log x - x)}$. **Q29.** $\log y = \sin x \log x \Rightarrow y' = x^{\sin x}(\cos x \log x + \frac{\sin x}{x})$. **Q30.** $A = \pi r^2 \Rightarrow \frac{dA}{dr} = 2\pi r = 2\pi(5) = 10\pi \text{ cm}^2/\text{cm}$. **Q31.** $y' = \frac{1}{x+\sqrt{x^2+a^2}} \cdot \left(1 + \frac{2x}{2\sqrt{x^2+a^2}}\right) = \frac{1}{\sqrt{x^2+a^2}}$. **Q32.** Intersection $y^3 = k$. Slopes: $m_1 = \frac{1}{2y}$,

$$m_2 = -\frac{k}{x^2} = -\frac{1}{y}. m_1 m_2 = -1 \implies -\frac{1}{2y^2} = -1 \implies 2y^2 = 1. \text{ Since } k = y^3 \implies k^2 = y^6 \implies 8k^2 = 8(1/8) = 1.$$

Q33. Let $u = x^x \implies u' = x^x(1 + \log x)$. Let

$$v = (\sin x)^{\log x} \implies \log v = \log x \log(\sin x) \implies v' = (\sin x)^{\log x} \left(\frac{\log(\sin x)}{x} + \log x \cot x \right)$$

. Add u' and v' . **Q34.** Slope $m = 3x^2 - 3$. At $x = 2$, $m = 9$. Tangent:

$$y - 7 = 9(x - 2) \implies 9x - y - 11 = 0. \text{ Normal:}$$

$$y - 7 = -\frac{1}{9}(x - 2) \implies x + 9y - 65 = 0. \text{ **Q35. } f(-4) = 0 = f(2). f \text{ is}**$$

continuous/differentiable. $f'(c) = 2c + 2 = 0 \implies c = -1 \in (-4, 2)$. Theorem verified. **Q36.**

$$\frac{dx}{d\theta} = a(1 - \cos \theta), \frac{dy}{d\theta} = -a \sin \theta. \frac{dy}{dx} = -\frac{\sin \theta}{1 - \cos \theta} = -\cot(\theta/2). \text{ At } \theta = \pi/3, \frac{dy}{dx} = -\sqrt{3}.$$

Q37. Let $x = \tan \theta$. $u = \tan^{-1}(\frac{\sec \theta - 1}{\tan \theta}) = \frac{\theta}{2} = \frac{1}{2} \tan^{-1} x$. Let $v = \tan^{-1} x$. $\frac{du}{dv} = \frac{1}{2}$. **Q38.**

$$y_1 = \frac{2 \sin^{-1} x}{\sqrt{1-x^2}} \implies y_1^2(1-x^2) = 4(\sin^{-1} x)^2 = 4y. \text{ Differentiating:}$$

$$2y_1 y_2(1-x^2) + y_1^2(-2x) = 4y_1 \implies (1-x^2)y_2 - xy_1 - 2 = 0. \text{ **Q39.}**$$

$f'(x) = 12(x-2)(x^2+2) = 0 \implies x = 2$. $f(0) = 25$, $f(2) = -39$, $f(3) = 16$. Absolute max is 25, absolute min is -39. **Q40.** $f'(x) = \cos x - \sin x = 0 \implies x = \pi/4, 5\pi/4$. Strictly increasing in $[0, \pi/4) \cup (5\pi/4, 2\pi]$. Strictly decreasing in $(\pi/4, 5\pi/4)$. **Q41.** Let

$$u = x^{\cos x} \implies u' = x^{\cos x} \left(\frac{\cos x}{x} - \sin x \log x \right). \text{ Let}$$

$$v = (\cos x)^x \implies v' = (\cos x)^x (\log(\cos x) - x \tan x). \text{ Add } u' \text{ and } v'. \text{ **Q42.}**$$

$$y' = \frac{1}{1+x} - \frac{4}{(2+x)^2} = \frac{x^2}{(1+x)(2+x)^2}. \text{ Since } y' \geq 0 \text{ for all } x > -1, \text{ it is strictly increasing.}$$

Q43. Let radius be r , height h , slant height l (constant). $V = \frac{1}{3}\pi r^2 h = \frac{1}{3}\pi(l^2 - h^2)h$.

$$V'(h) = \frac{1}{3}\pi(l^2 - 3h^2) = 0 \implies l^2 = 3h^2 \implies r^2 = 2h^2.$$

$$\tan \alpha = \frac{r}{h} = \sqrt{2} \implies \alpha = \tan^{-1} \sqrt{2}. V''(h) < 0 \text{ guarantees maximum. **Q44.}**$$

$f'(x) = 6(x-3)(x+2) = 0 \implies x = 3, -2$. Increasing in $(-\infty, -2) \cup (3, \infty)$. Decreasing in $(-2, 3)$. Local max at $(-2, 51)$, local min at $(3, -74)$. **Q45.** Let cylinder radius be r , height h .

$$R^2 = r^2 + (h/2)^2 \implies r^2 = R^2 - h^2/4. \text{ Volume } V = \pi r^2 h = \pi(R^2 h - h^3/4).$$

$$V'(h) = \pi(R^2 - 3h^2/4) = 0 \implies h = \frac{2R}{\sqrt{3}}. V''(h) < 0 \text{ validates maximum. **Q46.}** Base side$$

$$x, \text{ height } y. \text{ Area } c^2 = x^2 + 4xy \implies y = \frac{c^2 - x^2}{4x}. V = x^2 y = \frac{x(c^2 - x^2)}{4}.$$

$$V'(x) = \frac{c^2 - 3x^2}{4} = 0 \implies x = \frac{c}{\sqrt{3}}. V_{\max} = \frac{c^3}{6\sqrt{3}}. \text{ **Q47.}** Let square perimeter be } x, \text{ circle}$$

$$\text{circumference be } 28 - x. \text{ Area } A = \left(\frac{x}{4}\right)^2 + \pi\left(\frac{28-x}{2\pi}\right)^2.$$

$$A'(x) = 0 \implies \frac{x}{8} - \frac{28-x}{2\pi} = 0 \implies x = \frac{112}{\pi+4}. \text{ Thus, the wire should be cut at } \frac{112}{\pi+4} \text{ m (for square) and } \frac{28\pi}{\pi+4} \text{ m (for circle). **Q48.}**$$

$$f'(x) = 3x^3 - 24x^2 + 45x = 3x(x-3)(x-5) = 0 \implies x = 0, 3, 5.$$

$$f''(x) = 9x^2 - 48x + 45. f''(0) > 0 \text{ (Local Min at } x = 0), f''(3) < 0 \text{ (Local Max at } x = 3),$$

$$f''(5) > 0 \text{ (Local Min at } x = 5). \text{ **Q49. (i) } R(x) = (500 - x)(300 + x). \text{ (ii)}**$$

$$R'(x) = 200 - 2x = 0 \implies x = 100. \text{ (iii) } R''(x) = -2 < 0. \text{ Max revenue}$$

$$R(100) = 400 \times 400 = ₹1,60,000. \text{ **Q50. (i) Perimeter}**$$

$$P = x + 2y + \frac{\pi x}{2} = 10 \implies y = \frac{10 - x - \frac{\pi x}{2}}{2}. \text{ (ii) Area}$$

$$A(x) = xy + \frac{1}{2}\pi\left(\frac{x}{2}\right)^2 = 5x - \frac{x^2}{2} - \frac{\pi x^2}{8}. \text{ (iii) } A'(x) = 5 - x - \frac{\pi x}{4} = 0 \implies x = \frac{20}{\pi+4} \text{ m.}$$

$$\text{Using (i), } y = \frac{10}{\pi+4} \text{ m. } A''(x) < 0 \text{ confirms maximum.}$$